

## **Information as the main driver of the functioning of complex natural systems**

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### ***What hypothesis do you propose and why do you propose it?***

We propose that in complex ecological systems – composed of diverse biotic communities, containing at least three trophic levels (producers, consumers and decomposers), with their abiotic environment – information is the main driver of their performance, besides of the energy flow and matter cycling.

This information is qualitative in its nature, and is carried and stored in diverse complex compounds. Of these, hopanoids (a diverse group of polycyclic terpenoids produced by microorganisms, similar to steroids and also metabolically active, but much more persistent in the environment) are good candidates for the carriers of ecosystem memory, which is an often discussed concept, but with its mechanisms still remaining elusive. Also, aluminosilicates and organic polymers (natural – e.g. cellulose – or synthetic) may encode, store and transfer ecologically important information.

This endogenically produced information is reflecting the ecosystem history, particularly past events of intense perturbation (e.g. fires, floods), but may also be transferable on its carriers, and may be imposed by a conscious, direct human psychic influence, or consciously encoded on these carriers through specific symbols, which link the physical level with the semiotic one.

In biophysical sense, such information controls and directly regulates dynamic (eco)physiological processes and relates more to the derivatives of performance functions, than to stationary states of the system, indicating rather chaotic-based performance, than linear relationships within the system.

Our concept will lead to explaining the often observed patterns of diverse performance (in terms of metabolic processes and ecological succession) of physically and chemically similar ecosystems, different reactions on changes of the same kind (as the input of fertilizing or toxic substances), and even reported “unbelievable” results of some devoted intuitive gardeners. This is also an attempt to explain why singular, but intense perturbation events often leave strong traces in future system states. Our previous studies indicate changed metabolic rates and, consequently, affected final abundances and structures of the biotic communities under the influence of imposed external information, as well as sudden changes of some biophysical parameters (as redox potential), when such information is introduced.

### ***Why is your proposal of earth-shaking significance?***

According to the General Systems Theory, each complex system should be characterized by flow of energy, cycling (which may be partially open) of matter, and exchange of information. This is the main rule in the design of technical systems, and the role of information in social systems is also obvious. However, natural systems at all their levels of organization are usually described only quantitatively in terms of energy and matter flows, with their regulation left only to self-executed negative feedback loops.

Our concept is designed to show the relevance of Goodwin’s science of qualities – and not only quantitative descriptors – to functioning of complex ecological systems, and to relate to abandoned approaches of natural history in the description of performance of ecological systems.

Of particular importance there is the connection between natural systems and the conscious human influence, and the effect of the experimenter’s consciousness on the informational performance of complex ecological systems. In a long-term perspective, this approach leads to explaining the mutual links of natural and human components in the integrated common reality.

It is also putting the attention on diverse forms and carriers of information, other than genetic one, contained in DNA, while the recent reductionist paradigm limits the role of biological information up to the individual/population levels only. An emphasis on microbiomes, widely discussed recently, mainly in scales of internal functions of individual organisms, will lead to broadening the microbiome concept on larger systems, with microbial roles in production of information and translating it into metabolic processes of the whole system.

### ***Briefly, how will you test your hypothesis?***

Experiments will be carried in aquatic microcosms (enabling better performance control and as it is expected that information is carried through dynamic patterns of water structures), containing diverse communities of producers (algae/macrophytes), consumers (invertebrates or small fish) and decomposers (bacteria/fungi), and

the source of particular organic matter (organic sediments or specific tree leaves). All basic rules of experimental design (controls, repetitions enabling statistical analyses, the same initial descriptors except the studied one, etc.) will be maintained. Intense perturbation events will be produced to execute changes in information; subsequently, hopanoids will be extracted, determined biochemically with chromatographic methods, and transferred (in minute quantities) to previously non-stressed microcosms.

The performance of microcosms will be described in terms of ecophysiological rates (production/release of oxygen, dissolved organics or specific nutrients), physicochemical parameters of water, and changes of biotic community structure and the abundance of its dominants. Metabolic patterns of particular substances will be determined as their concentrations with appropriate chemical analytical methods, as well with bioassessment methodology, as there is expected that the same quantities may produce different biological effects according to information applied. Additionally some alternative descriptors will be applied (as crystallization patterns of salts in dried water drops, or radiesthetic biometer readings).

Intentional information will be encoded on natural aluminosilicates using an unique method created in the Subtle Energies' Laboratory and tested from over 12 years. The main procedure is based on a specific programming language to encode the aluminosilicates. During our studies we have proven that loading information into matter is possible when consciousness of the operator – at least within a part of the process – became one of elements of the programming system. Without this element all the programming system (i.e. program language and specific hardware) are non-productive, there will be no encoded matter at the end of process. The operator must know rules how to include his/her consciousness with its non-determined creative possibilities into technical (algorithmic) process of programming. This requirements may be fulfilled by using symbols as words of the programming language. The symbol performs many functions: it is an attractor defining a specific “field of meanings” (as the attractor creates the fractal, acc. to the fractal theory), it includes both physical and semiotic essences), so it is very useful medium connecting active consciousness and unconditional Laws of Nature.

After applying consciousness into programming process, we may use other specific methods for information transfer, as the collection of magnetic footprints, light-powered transfer, etc. In the next stage, some similar experiments and measurements will be performed in larger, natural ecosystems, including e.g. ponds formed in gravel pits (considered as “landscapes without a memory”), as well as successional mature, stable aquatic bodies (expected to be regulated by memory-based internal information). The unique property of information activity is that the result are visible only in large systems, usually indirectly, as changing the parameters of the system not easily associated with the primary cause. That's the reason why we cannot use traditional reductionist methodology to investigate the information phenomena. Instead we propose to observe changes not directly connected to inserted information, but relating to the whole system. The results often may appear within the system elements, where we do not expect them according to any traditional cognitive model.

Our research formula, although focused on particular information roles, is open for other relevant and complimentary concepts and experimental designs, and allowing for adaptive management (including the use of pre-programmed worksheets, calculating and visualizing the results immediately when raw data are entered).

### ***Which field of science or group of scientists does your proposal challenge?***

The main disciplines challenged are the ecosystem- or landscape-level ecology and biogeochemistry. Their evidence-based models may work well without information considered, based only on quantified energy and matter flows, particularly when the evidence comes from simple experimental designs, based on monospecific cultures, or interactions of few species only. This approach resembles ecosystem models from several decades ago, when adding to the plant and animal data also the obtainable data on microbial components (based those days on some 1% of lab-cultivable bacteria) often worsened the model performance. Recently, we cannot even imagine ecosystem models without microbial functions.

If traditional scientists only allow their acceptance of psychotronic effects, they usually automatically limit them to the human mind, or higher animals as the latest, but never for whole ecosystems, with microorganisms included. Therefore we challenge also the paradigm that ecosystems may be natural, totally independent of humans, or anthropogenically (i.e. physically) modified, and there is no room for effects resulting from the experimenter's action, or for non-physical and indirect influences of any humans. Our concept proposes the integration of natural and human components within one system.

We also challenge the view that information – if only considered in ecological systems – remains limited to this encoded and transferred with DNA as the genetic information only.